A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The atmosphere: energy processes (UV, IR, visible light) - Set 1

Mark scheme

AQA spec 3.2.1.1; The physical environment; textbook pp. 86-89

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** CO_2 rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again **1** During spring / summer, photosynthesis by plants (growing season) removes CO_2 faster than respiration releases it, so the concentration falls **1** In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO_2 rises **1** The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean – more land / vegetation in the northern hemisphere makes the cycle large **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) **1** So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **denitrification** → nitrate converted back to nitrogen gas • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
01.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) **1** Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) **1** Atmospheric pressure allows liquid water to exist **1** Distributes heat from the tropics to higher latitudes by winds **1** Transports water vapour to areas that would otherwise be dry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Troposphere **1** Stratosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 + 5 = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • A bare wrong answer with no working. • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	1
02.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 UV:** short wavelength – mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium – converted to chemical energy in these reactions – little reaches the surface **2 Visible light:** transmitted through the atmosphere easily (some reflected by clouds / dust) – absorbed by the Earth's surface and converted to heat – some absorbed by chlorophyll and converted to chemical energy in photosynthesis **2 IR:** emitted by the warm surface (long-wave) – absorbed by greenhouse gases (CO2, water vapour, methane) and converted to heat – the warm atmosphere re-emits IR to the surface and reduces conduction losses – some escapes to space **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down **1** The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy – only the net loss (70%) balances the absorbed solar energy at equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
02.4	1 mark for the point below. **1** The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead **1** The warm surface heats the air above – warm winds (eg south-westerlies) carry heat energy to higher latitudes **1** Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe **1** Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes **OR** transports water to areas that would be dry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun • **latent heat** → the energy absorbed or released when water changes state	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe **1** Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Atmospheric pressure controls how easily water molecules evaporate / escape from the surface **1** At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Nitrogen **1** Oxygen **1** Carbon dioxide **1** Argon / neon / krypton / xenon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 15 - (-18) = 33 °C **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	1
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** Visible light passes easily through the atmosphere and is absorbed by the surface, which warms **1** The warm surface emits (long-wave / far) infrared radiation **1** Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat **1** The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	4
05.3	1 mark for the point below. **1** Any gas that is better at absorbing infrared radiation than the average for the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	1
05.4	**1 mark each** 1 mark for each of the 2 points below. **1 Arriving:** mainly ultraviolet, visible light and near infrared (short wavelength) – leaving: mainly long-wavelength (far) infrared **1** Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

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Question	Marking guidance	Marks
05.5	**1 mark each** 1 mark for each of the 2 points below. **1** The energy and the processes it drives control climate, ocean currents and the hydrological cycle **1** These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
06.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

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Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1
06.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

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Question	Marking guidance	Marks
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

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